hdoc day 51 by trishil chauhan

Problem statement

Read the size and elements of a variable-length integer array. Pass its address to **next_greater()**. Using nested loops, find the nearest strictly greater element to the right of each element, or print -1 if none exists. Display the answers in original order, separated by commas.

Complete C program

```
#include <stdio.h>

void next_greater(const int *arr, int n) {
    int i, j, next;

    printf("Next greater elements: ");
    for (i = 0; i < n; i++) {
        next = -1;
        for (j = i + 1; j < n; j++) {
            if (*(arr + j) > *(arr + i)) {
                next = *(arr + j);
                break;
            }
        }
        printf("%d", next);
        if (i < n - 1) {
            printf(", ");
        }
    }
    printf("\n");
}

int main(void) {
    int n, i;

    printf("Enter array size: ");
    if (scanf("%d", &n) != 1 || n <= 0) {
        printf("Invalid array size.\n");
        return 1;
    }

    int arr[n];
    printf("Enter %d elements: ", n);
    for (i = 0; i < n; i++) {
        if (scanf("%d", &arr[i]) != 1) {
            printf("Invalid input.\n");
            return 1;
        }
    }

    next_greater(arr, n);
    return 0;
}
```

Sample output

```
Enter array size: 5
Enter 5 elements: 4 5 2 10 8
```

```
Next greater elements: 5, 10, 10, -1, -1
```

Explanation

For 4, the nearest greater value to the right is 5. For 5 and 2, it is 10. Neither 10 nor 8 has a greater element to its right, so both produce -1. The inner loop stops at the first greater value, ensuring the nearest qualifying element is selected.

Complexity

Time: $O(n^2)$ in the worst case because of the nested loops. Auxiliary space: O(1), excluding the input variable-length array.